

ePG Introduction to Machine Learning Quiz 1

Solutions

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Solutions

1. Constrained Optimization

- (a) Convex: Yes. Hessian is positive definite.
- (b) Feasible region:

$$x_1 + 2x_2 \leq -1$$

- (c) Lagrangian:

$$\mathcal{L} = x_1^2 + x_2^2 + \lambda(x_1 + 2x_2 + 1)$$

$$x_1 = -\frac{\lambda}{2}, \quad x_2 = -\lambda$$

$$-\frac{\lambda}{2} - 2\lambda = -1 \Rightarrow \lambda = \frac{2}{5}$$

$$x^* = \left(-\frac{1}{5}, -\frac{2}{5}\right)$$

2. Linear Algebra

- (a) Not linearly independent (3 vectors in $\mathbb{R}^2$).
- (b) Span is $\mathbb{R}^2$.

3. SVM

- (a) **Max-margin separability:** No.

The data is not linearly separable. The crosses lie in the inner region while the dots form an outer ring, so no straight line can separate the two classes.

- (b) **Kernel SVM:** Yes.

A kernel SVM can separate this data by mapping it to a higher-dimensional feature space where a linear separator exists.

A kernel SVM implicitly maps the data to a higher-dimensional feature space $\phi : \mathbb{R}^2 \rightarrow \mathcal{H}$ via a kernel

$$K(x, x') = \langle \phi(x), \phi(x') \rangle.$$

In the higher-dimensional space, the data may become linearly separable even if it is not in $\mathbb{R}^2$.

Common kernels include the RBF (Gaussian) kernel:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2),$$

which can separate such circular (non-linear) patterns.

4. Soft-Margin SVM

(a) **Fraction correctly classified:**

In soft-margin SVM:

- $\theta_i = 0 \Rightarrow$ correctly classified (outside or on margin)
- $0 < \theta_i \leq 1 \Rightarrow$ correctly classified but inside margin
- $\theta_i > 1 \Rightarrow$ misclassified

From the table:

$$\theta_5 = 1.5 > 1 \Rightarrow \text{misclassified}$$

All others have $\theta_i = 0$.

$$\text{Correctly classified} = 5/6$$

(b) **Support vectors:**

Support vectors satisfy $\alpha_i^* > 0$.

From table:

$$i = 3, 4, 5$$

(c) **Removing sample 5:**

Yes, the classifier will change.

Sample 5 has $\alpha_5^* = 1 > 0$, so it is a support vector and contributes to

$$w^* = \sum_i \alpha_i^* y_i x_i$$

Removing it changes w^* and hence the decision boundary.

(d) **Removing sample 6:**

No, the classifier will not change.

Sample 6 has $\alpha_6^* = 0$, so it does not contribute to w^* .

5. Probability

(a)

$$\left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

(b)

$$E = 7$$

6. Linear Regression

We fit the model $\hat{y} = \beta_0 + \beta_1 x$ using ordinary least squares.

Step 1: Summary statistics

$$n = 3, \quad \bar{x} = \frac{1 + 3 + 5}{3} = 3, \quad \bar{y} = \frac{40 + 65 + 75}{3} = 60$$

Step 2: Compute slope

$$\begin{aligned} S_{xx} &= (1 - 3)^2 + (3 - 3)^2 + (5 - 3)^2 = 4 + 0 + 4 = 8 \\ S_{xy} &= (1 - 3)(40 - 60) + (3 - 3)(65 - 60) + (5 - 3)(75 - 60) \\ &= (-2)(-20) + 0 + (2)(15) = 40 + 30 = 70 \end{aligned}$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{70}{8} = 8.75$$

Step 3: Compute intercept

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 60 - 8.75 \times 3 = 33.75$$

Model:

$$\hat{y} = 33.75 + 8.75x$$

Prediction at $x = 0$:

$$\hat{y} = 33.75$$

7. Gradient Descent

$$f(x) = x^4 - 2x^2, \quad f'(x) = 4x^3 - 4x = 4x(x^2 - 1)$$

Update rule:

$$x_{t+1} = x_t - \eta f'(x_t), \quad \eta = 0.1$$

(a) $x_0 = 0$

$$f'(0) = 0$$

$$x_1 = 0, \quad x_2 = 0$$

Gradient is zero, so the algorithm does not move.

(b) $x_0 = 0.1$

$$f'(0.1) = 4(0.1)(0.1^2 - 1) = -0.396$$

$$x_1 = 0.1 - 0.1(-0.396) = 0.1396$$

$$f'(0.1396) \approx -0.5475$$

$$x_2 = 0.1396 - 0.1(-0.5475) \approx 0.1944$$

(c) $x_0 = -0.1$

$$f'(-0.1) = 0.396$$

$$x_1 = -0.1 - 0.1(0.396) = -0.1396$$

$$f'(-0.1396) \approx 0.5475$$

$$x_2 = -0.1396 - 0.1(0.5475) \approx -0.1944$$

(d) **Interpretation**

Critical points:

$$f'(x) = 0 \Rightarrow x = -1, 0, 1$$

Second derivative:

$$f''(x) = 12x^2 - 4$$

$$f''(0) = -4 < 0 \Rightarrow \text{local maximum}$$

$$f''(\pm 1) = 8 > 0 \Rightarrow \text{local minima}$$

Conclusion:

- Gradient descent stalls at $x = 0$ because $f'(0) = 0$, even though it is a maximum.
- Starting from $x_0 > 0$, iterates move toward $+1$.
- Starting from $x_0 < 0$, iterates move toward -1 .